

1. Write an AbstractFactory interface to produce shoes and clothing. The brands include Nike, Anta, and LiNing. Write a shoe interface and a clothing interface, which have a show() function to show its information of brands. Write a Test class to create different brands of shoes and clothing.

AbstractFactory

```
package experiment5_1;

interface AbstractFactory {

    Clothing createClothing();

    Shoe createShoe();

}
```

AntaClothing

```
package experiment5_1;

public class AntaClothing implements Clothing{

    @Override

    public void show() {

        System.out.println("This is an Anta clothing.");

    }

}
```

AntaFactory

```
package experiment5_1;

public class AntaFactory implements AbstractFactory{
```

```
@Override  
  
public Clothing createClothing() {  
    return new AntaClothing();  
}  
  
@Override  
  
public Shoe createShoe() {  
    return new AntaShoe();  
}  
}
```

AntaShoe

```
package experiment5_1;  
  
public class AntaShoe implements Shoe{  
    @Override  
    public void show() {  
        System.out.println("These are Anta shoes");  
    }  
}
```

Clothing

```
package experiment5_1;
```

```
interface Clothing {  
    void show();  
}
```

LiNingClothing

```
package experiment5_1;  
  
public class LiNingClothing implements Clothing{  
    @Override  
    public void show() {  
        System.out.println("This is a LiNing clothing.");  
    }  
}
```

LiNingFactory

```
package experiment5_1;  
  
public class LiNingFactory implements AbstractFactory{  
    @Override  
    public Clothing createClothing() {  
        return new LiNingClothing();  
    }  
  
    @Override
```

```
    public Shoe createShoe() {  
        return new LiNingShoe();  
    }  
}
```

LiNingShoe

```
package experiment5_1;  
  
public class LiNingShoe implements Shoe{  
    @Override  
    public void show() {  
        System.out.println("These are LiNing shoes.");  
    }  
}
```

NikeClothing

```
package experiment5_1;  
  
public class NikeClothing implements Clothing{  
    @Override  
    public void show() {  
        System.out.println("This is a Nike clothing.");  
    }  
}
```

NikeFactory

```
package experiment5_1;

public class NikeFactory implements AbstractFactory{

    @Override

    public Clothing createClothing() {

        return new NikeClothing();

    }

    @Override

    public Shoe createShoe() {

        return new NikeShoe();

    }

}
```

NikeShoe

```
package experiment5_1;

public class NikeShoe implements Shoe{

    @Override

    public void show() {

        System.out.println("These are Nike shoes.");

    }

}
```

```
}  
}
```

Shoe

```
package experiment5_1;  
  
interface Shoe {  
    void show();  
}
```

Test

```
package experiment5_1;  
  
public class Test {  
    public static void main(String[] args) {  
        System.out.println("***Anta Factory***");  
        AbstractFactory abstractFactory1 = new  
AntaFactory();  
        Clothing clothing1 =  
abstractFactory1.createClothing();  
        Shoe shoe1 = abstractFactory1.createShoe();  
        clothing1.show();  
        shoe1.show();  
    }  
}
```

```
        System.out.println("***Nike Factory***");

        AbstractFactory abstractFactory2 = new
NikeFactory();

        Clothing clothing2 =
abstractFactory2.createClothing();

        Shoe shoe2 = abstractFactory2.createShoe();

        clothing2.show();

        shoe2.show();


        System.out.println("***LiNing Factory***");

        AbstractFactory abstractFactory3 = new
LiNingFactory();

        Clothing clothing3 =
abstractFactory3.createClothing();

        Shoe shoe3 = abstractFactory3.createShoe();

        clothing3.show();

        shoe3.show();

    }
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\Int
***Anta Factory***
This is an Anta clothing.
These are Anta shoes
***Nike Factory***
This is a Nike clothing.
These are Nike shoes.
***LiNing Factory***
This is a LiNing clothing.
These are LiNing shoes.

Process finished with exit code 0
```